Q1

Student ID → xx ... 4 3 2 1 (last four digits)

$C = (3+2+1) + 1 = 7$

(3rd, 2nd, 1st digits; 1 fixed)

$C =$? Write only the answer, no calculations!!!

Q2

For a silicon crystal doped with Boron, what must be N_A at $T=300K$ if the electron concentration drops to $\frac{n_i}{100 \times C}$.

$n_i = 1.5 \times 10^{10}$ carriers/cm³ @ 300 K ↑ from Q1

Hint : $n_i^2 = n_p \cdot n_e$

$n_e = n_i/100.C = n_i/100.7$ $n_p = N_A$

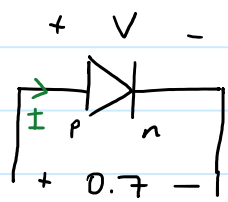
$n_i^2 = N_A \cdot \frac{n_i}{700} \Rightarrow N_A = 700 * 1.5 \times 10^{10}$

$= 1.05 \times 10^{13}$ atoms/cm³

Q3

from Q1

For a silicon diode $I_S = C \times 10^{-16}$ A. Calculate the current if 0.7V is applied across it at 300K.



Hint: $I = I_S (e^{V/V_T} - 1)$ & $V_T \cong 25mV$ at 300K.

Just apply the formula

$$I = C \cdot 10^{-16} \left(e^{V/0.025} - 1\right)$$

$$= 7 \cdot 10^{-16} \left(e^{0.7/0.025} - 1\right)$$

$$= 0.001\,A = 1mA$$